

POWER MANAGEMENT STUDY

A Strategic Framework for Renewable Integration, Energy Storage, and Demand Optimization



PROBLEM STATEMENT : THE GRID IMBALANCE

- Sri Lanka's solar PV capacity is expanding, creating a surplus during daytime off-peak hours.
- Due to the intermittent nature of solar and a lack of bulk energy storage, grid integration remains highly challenging.
- During the critical evening peak (19:15), solar generation drops to zero.
- Consequently, the grid is forced to rely heavily on expensive, carbon-intensive thermal power plants to meet night demand.



FIGURE 01:SOLAR POWER GENERATION

THE "DUCK CURVE" CHALLENGE

DAYTIME SOLAR SURGE

As solar energy production peaks in the middle of the day, it significantly reduces the net load on conventional generators.

Without proper storage, this surplus energy cannot be fully utilized, leading to potential curtailment and grid instability.

EVENING PEAK DEMAND

As the sun sets, solar generation rapidly declines just as residential power demand spikes. This requires other power sources, primarily thermal and major hydro, to ramp up extremely quickly to cover the massive supply deficit.

THE PROPOSED SOLUTION: PUMPED HYDRO STORAGE

To solve the solar intermittency issue without relying on expensive chemical batteries, we propose utilizing Sri Lanka's extensive existing hydroelectric infrastructure.

During the day, surplus solar energy can be used to pump water from a lower reservoir back up to a higher reservoir. During the night peak, this stored water is released through turbines to generate electricity, effectively acting as a massive grid-scale battery.

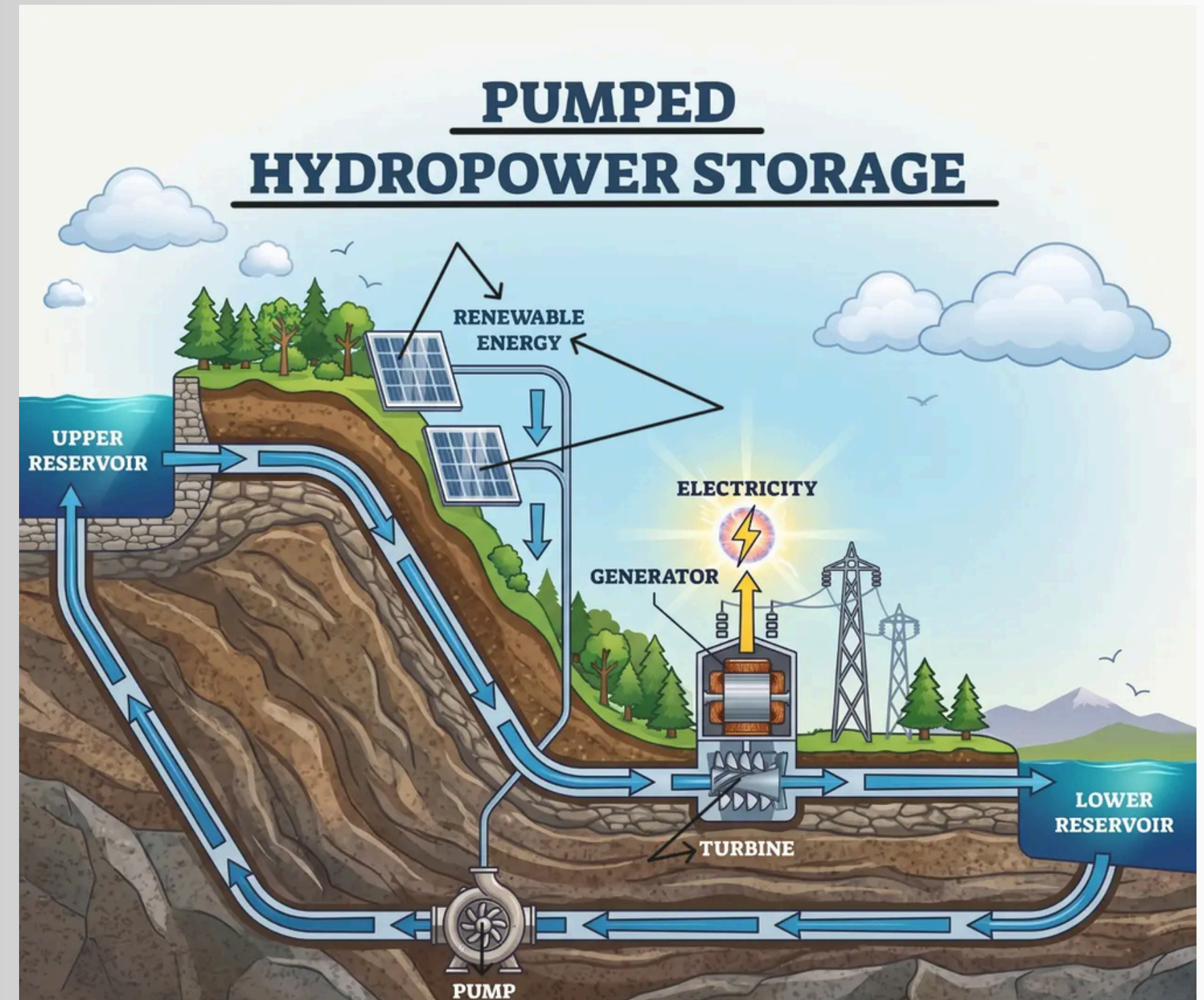
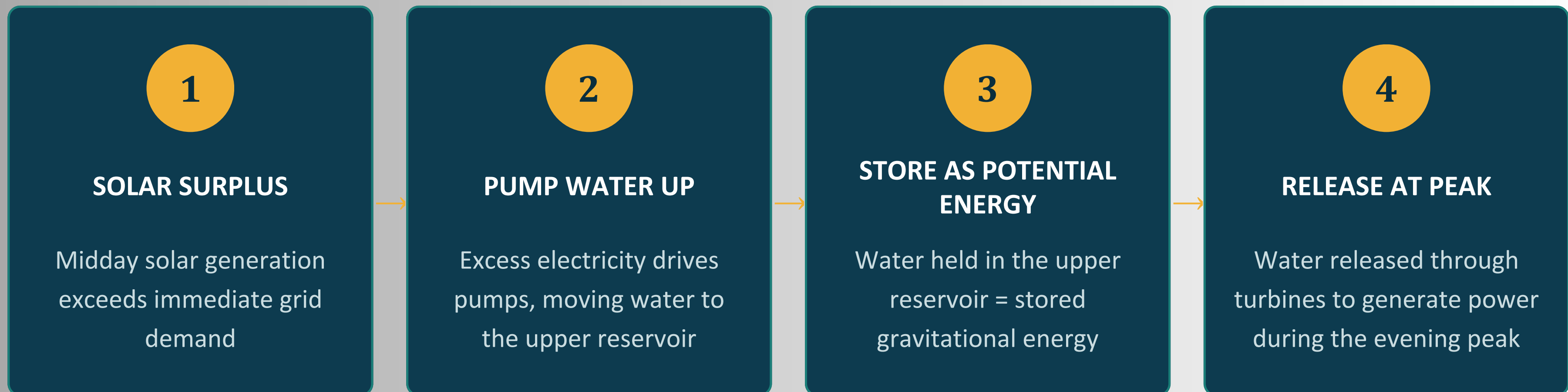


FIGURE 02: PUMPED HYDRO STORAGE

THE PROPOSED SOLUTION: PUMPED HYDRO STORAGE



WHY GRAVITY INSTEAD OF CHEMISTRY?

Criterion	Pumped-Storage Hydro (PSH)	Chemical Battery
Lifespan	40–100 years (civil works, refurbishable)	8–15 years before major degradation
Maintenance cost	Low — mostly mechanical/electrical upkeep	High — cell monitoring, thermal management
Replacement	Rare (turbines/pumps refurbished, not replaced)	Frequent — full pack replacement over asset life
Existing infrastructure	Leverages reservoirs already built	Requires new sites, land, and supply chains
Storage duration	Hours to days — suited to daily cycling	Typically shorter duration, cost rises with capacity

TABLE O1: GRAVITY BATTERY AND ELECTROCHEMICAL BATTERY COMPARISON

DAILY GENERATION MIX

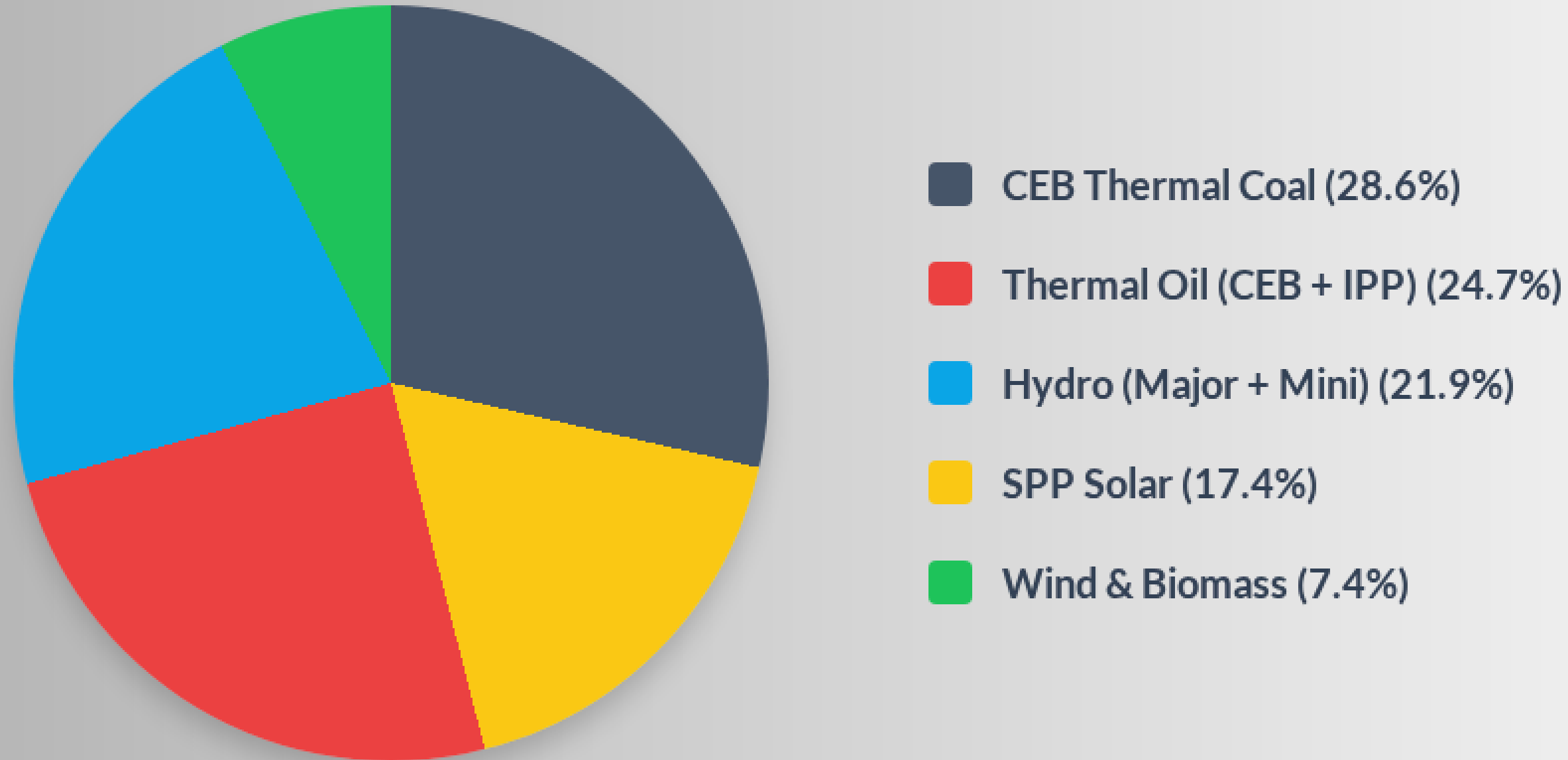


FIGURE 03: DAILY POWER GENERATION

Daily generation (Total: 57.19 GWh) relies heavily on Coal and Thermal Oil (over 53% combined), despite significant contributions from Solar during daytime hours.

The Daily Mismatch: Solar Surplus vs. Evening Peak

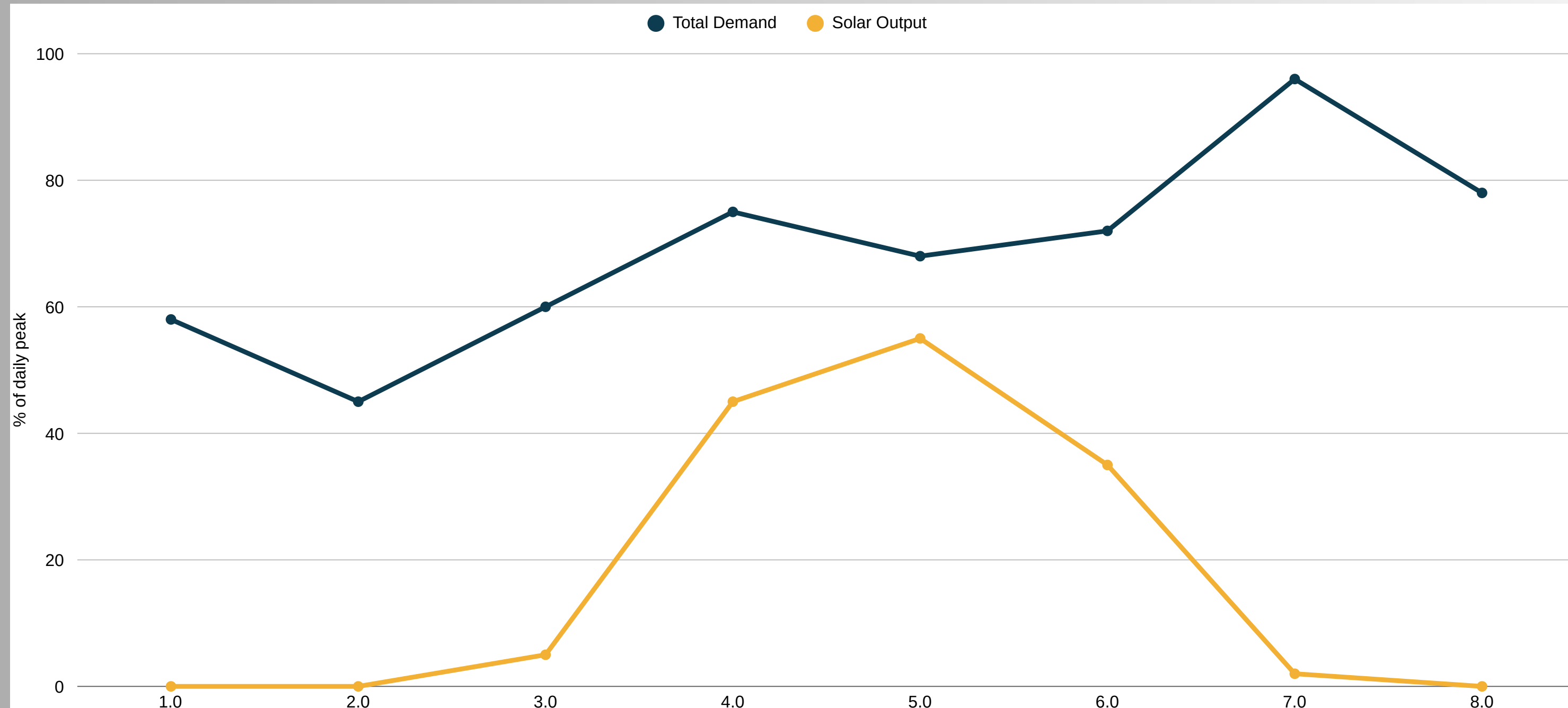


FIGURE 04: SOLAR SURPLUS VS.EVENING PEAK

Solar output collapses to near-zero exactly as demand climbs toward the 19:00–20:00 night peak. Midday solar surplus, if it could be stored, would otherwise directly offset the thermal ramp-up seen every evening in the generation curve.

STRATEGIC PROJECT ROADMAP



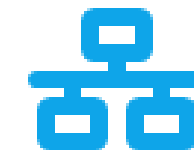
Hydro Assessment

Investigating existing conventional hydro plants and reservoir geometries to determine mechanical potential for pumped storage retrofiting.



Demand Forecasting

Analyzing historical demand patterns and utilizing data-driven models to predict future load dynamics and precise storage capacity requirements.



Centralized Planning

Synthesizing resource assessments and dispatch strategies into a cohesive framework to minimize reliance on thermal generation.

WASTE-TO-ENERGY SYSTEMS



FIGURE 05 : A GARBAGE POWER PLANT

Localized Power Generation

As a supplementary solution within our centralized framework, we are proposing the implementation of small-capacity garbage power plants (Waste-to-Energy).

These facilities address dual challenges: mitigating municipal solid waste accumulation and providing a consistent, decentralized baseload power supply to the grid, further reducing reliance on imported fossil fuels.

EXPECTED OUTCOMES

- 1 Assessment of existing renewable power resources across the grid
- 2 Identification of potential pumped-storage opportunities in the hydro network
- 3 A centralized framework for renewable power planning and management
- 4 Actionable recommendations for improving renewable energy utilization
- 5 A data-informed foundation for future detailed technical & engineering studies

POWER MANAGEMENT STUDY PLAN

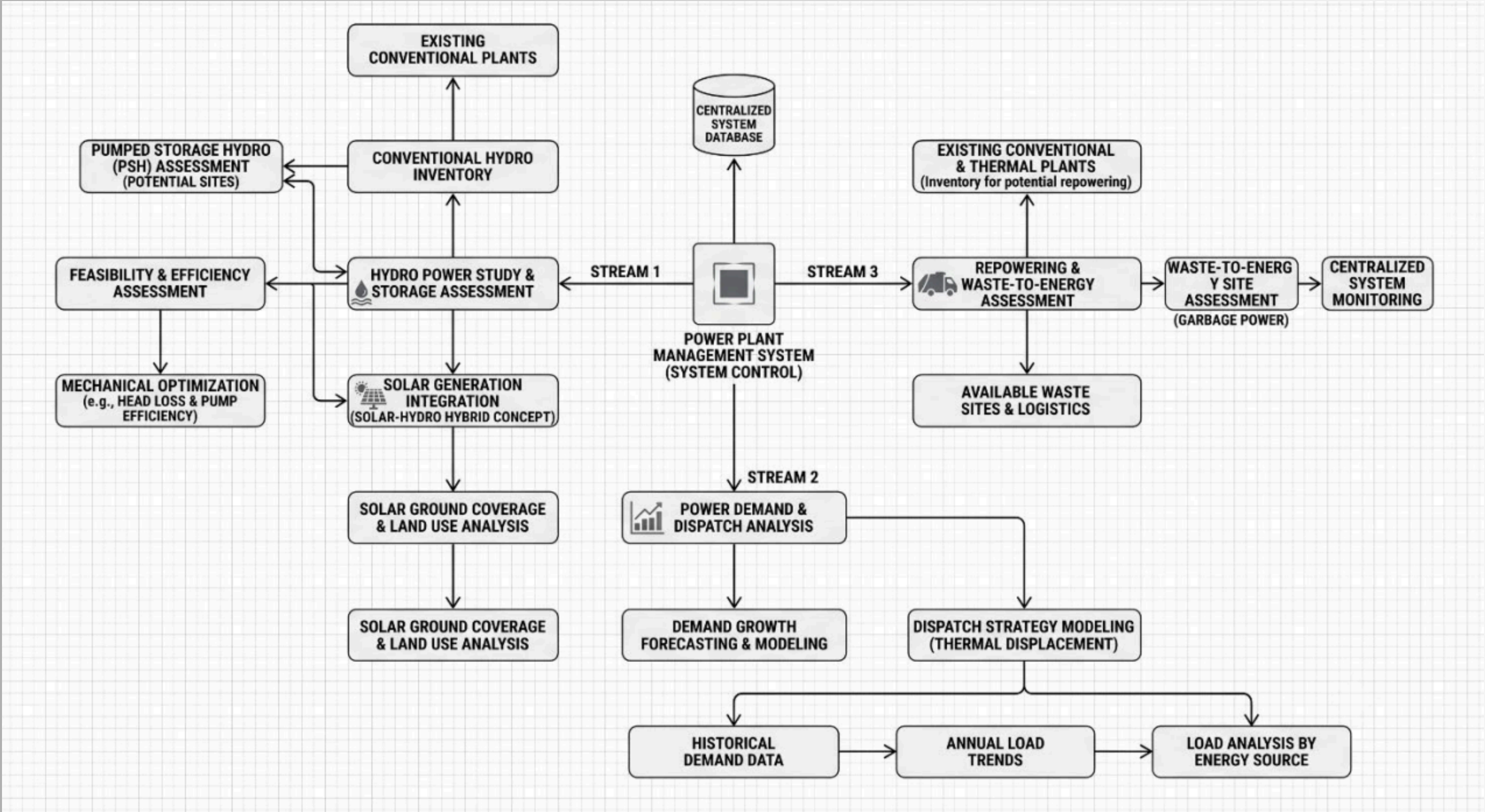


FIGURE 06 : PROJECT STUDY PLAN

Questions?

THANK YOU!

University of Peradeniya | Department of Mechanical Engineering | E22 Batch